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Causal Transformers for Energy Markets: Discovering Causal Dependencies in European Electricity Time Series

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Abstract

Transformer-based temporal causal discovery can identify directed, lagged relationships in multivariate time series. However, real-world electricity-market data have no known ground-truth causal graph against which these relationships can be evaluated. They must therefore be treated as candidate causal hypotheses rather than causal facts. This thesis applies CausalFormer, a transformer-based causal discovery model, to Germany-Luxembourg day-ahead electricity-market and weather data from 2019 to 2021, covering price, load, wind generation, solar generation, temperature, wind speed, and solar radiation. The proposed edges are assessed using a multi-stage validation framework that combines synthetic benchmarking, statistical triangulation, falsification tests, and robustness checks. On synthetic data, CausalFormer achieves moderate graph recovery but weaker lag recovery. On the real data, most cross-variable candidate edges fail at least one falsification test, while several highly stable edges are physically implausible, showing that stability alone is not sufficient evidence of causality. The strongest empirically supported dependency is temperature $\rightarrow$ load. The main contribution of this thesis is the validation framework rather than any single discovered edge. The results show that transformer-based causal discovery can be useful for generating causal hypotheses, but its outputs require independent validation and domain knowledge before they can be interpreted causally.

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1. Introduction

1.1. Background and Motivation

Electricity markets are shaped by weather, demand cycles, generation mix, and policy. Operators, traders, and regulators benefit from understanding how these factors relate, and whether one variable drives another or both respond to a common cause. Causal discovery methods attempt to infer such relationships from observational data, but the task is difficult. Controlled experiments are rarely feasible, and observational data alone cannot uniquely identify the causal structure (Peters, Janzing, and Schölkopf, 2017).

Traditional statistical approaches such as Granger causality and conditional-independence tests have been used in energy economics for decades. More recently, deep learning models, particularly transformers, have been applied to temporal causal discovery. These models are designed to capture nonlinear dependencies and long-range interactions, and their attention mechanisms offer a form of interpretability. They indicate which past observations the model considers relevant when predicting each variable. CausalFormer (Kong et al., 2024) is one such model, designed to output directed, lagged causal graphs from multivariate time series.

Model-derived relevance scores, however, are not causal evidence on their own. A model may assign importance to a variable because it is predictively useful, not because it is a genuine cause. Spurious correlations, confounders, and autoregressive patterns can all produce high relevance scores without reflecting true causal mechanisms. When CausalFormer or similar models are applied to real-world data, the resulting graphs are proposals, not ground truth. In such settings, there is no observable reference against which to verify them.

This creates a validation problem. Without synthetic benchmarks with known structure, or independent statistical tests, or robustness checks, there is no principled way to distinguish credible edges from artefacts. A transformer may propose dozens of causal relationships. Some may be stable and physically interpretable, while others may be

fragile, sensitive to preprocessing, or inconsistent with domain knowledge. Treating all model outputs as equally credible is not scientifically appropriate.

This thesis addresses the validation problem. It applies CausalFormer to electricity-market time series from the Germany-Luxembourg bidding zone, extracts directed and lagged candidate edges, and subjects them to a multi-stage validation framework. The framework includes synthetic benchmarks, statistical triangulation with PCMCI+ and Granger causality, falsification tests that corrupt temporal structure, and robustness checks across bootstrap resamples and seasonal regimes. The goal is not to prove that any edge is causal, but to assess which edges are credible enough to warrant interpretation, and which should be treated with scepticism.

1.2. Research Questions

This thesis is guided by two research questions:

RQ1: To what extent are CausalFormer-proposed edges statistically credible and robust under independent validation?

RQ2: Which proposed dependencies receive support in the DE-LU data, and how can they be interpreted?

RQ1 concerns method validation and asks whether CausalFormer’s outputs can be trusted. RQ2 concerns substantive findings and asks what the accepted edges reveal about electricity-market dependencies.

1.3. Contribution

This thesis designs and applies a multi-stage validation framework for transformer-generated causal edges in energy-market time series. The contribution is methodological rather than architectural. No new causal discovery model is proposed. Instead, the thesis evaluates what evidence remains after CausalFormer’s candidate edges are tested against independent statistical methods, falsification procedures, and robustness criteria.

The empirical application focuses on the Germany-Luxembourg day-ahead electricity market using hourly data from 2019 to 2021. CausalFormer proposes a set of cross-variable candidate edges, which the validation framework classifies into tiers based on their independent support. The main finding is that most candidates fail at least one validation stage, while only a small subset can be considered credible causal hypotheses.

Stability alone does not indicate credibility. Several edges that are highly stable under bootstrap resampling are flagged as spurious by falsification tests.

The thesis contributes to the growing literature on interpretable machine learning for causal inference by demonstrating that transformer outputs require systematic validation, and by providing a practical framework for conducting such validation in observational time-series settings.

All code, data, and results underlying this thesis are available in a public repository at <https://github.com/kalirole/causalformer-for-energy-markets>. The repository contains the preprocessing pipeline, the experiment configurations, the validation framework, and the scripts that generate the reported tables and figures, so that every result presented here can be reproduced.

1.4. Thesis Structure

Chapter 2 reviews causal discovery in time series, transformer-based approaches, and prior work on electricity-market causal analysis. Chapter 3 describes the data, preprocessing, CausalFormer pipeline, and validation framework. Chapter 4 presents the synthetic benchmark performance and the validation outcomes for the DE-LU candidate edges. Chapter 5 interprets the findings, discusses limitations, and considers implications. Chapter 6 summarizes the answers to the research questions and offers final remarks.

2. Background

2.1. Causal Discovery in Time Series Data

Most data analysis looks at statistical association: which variables move together, and how well one predicts another. Causal reasoning asks a harder question: how a system would react if one of its variables were changed. Two variables can be strongly correlated without one causing the other, for example when both follow a common cause. A model built on such correlations can predict one variable from the others, but it cannot say what would happen under an intervention. Causal discovery therefore tries to recover which variables actually affect which, rather than only a model that predicts well (Pearl, 2000; Peters, Janzing, and Schölkopf, 2017).

The most direct way to identify causal relationships is through controlled experiments. In many domains, however, including electricity markets, such experiments are impossible, prohibitively expensive, or unethical. Researchers therefore have to rely on observations. These alone cannot uniquely identify the underlying causal structure because multiple causal graphs can generate the same statistical dependencies. Additional assumptions are therefore required to distinguish between them. (Peters, Janzing, and Schölkopf, 2017). This is known as the identifiability problem, and is the core difficulty of the field (Glymour, K. Zhang, and Spirtes, 2019). It becomes even harder with time-series data.

Time-series data both help and hinder causal discovery. On the one hand, time gives a natural order: a cause must come before its effect, which rules out some relations that cross-sectional data cannot rule out (Runge, Gerhardus, et al., 2023). On the other hand, the features that make time series realistic also make them hard to analyse (Runge, Gerhardus, et al., 2023; Gong et al., 2023). Consecutive observations are correlated rather than independent. Autocorrelation is often so strong that a variable’s own past predicts its present almost perfectly, which hides the effect of other variables. In electricity markets, causal effects do not always occur at a known or fixed delay. One variable may affect another after one hour, several hours, or through several delayed responses. Some

effects may also happen within the same observation interval, so temporal order alone is not always enough to determine the direction of causality. In addition, relationships can be nonlinear or indirect, meaning that one variable may influence another through a third variable. Unobserved common causes, or hidden confounders, create associations that conditioning on the measured variables cannot remove, and measurement noise adds further uncertainty (Bartlett, 1946). Finally, most real systems are non-stationary: their statistical properties change over time, so a relationship found in one period may not hold in another. Any method for observational time series must deal with some mix of these problems, and the assumptions it makes about them decide what it can recover (Runge, 2020).

These challenges, together with the growing availability of observational time series across science, industry, and finance, have led to a large body of research on temporal causal discovery, with methods trading off different assumptions, computational cost, and modelling power (Gong et al., 2023; Runge, 2020). The next section reviews the main methodological families and their strengths and limitations.

2.2. Statistical Approaches to Temporal Causal Discovery

Temporal causal discovery builds on a broad set of classical methods, many of which were developed before the recent rise of deep learning. Since temporal datasets differ in their degree of autocorrelation, non-stationarity, noise, and potential confounding, causal structure cannot be recovered by a single universally applicable strategy. Instead, methods differ in the assumptions they impose on the data-generating process and in the criterion used to identify causal relationships (Peters, Janzing, and Schölkopf, 2017). The literature commonly distinguishes between constraint-based methods, score-based methods, and functional causal models (Gong et al., 2023; Runge, Gerhardus, et al., 2023). These families respectively rely on conditional independence tests, score-optimization principles, or assumptions about causal mechanisms and noise distributions. Each family is described below by its inference principle, representative methods, strengths, and limitations.

Constraint-based methods infer structure from the conditional dependencies and independencies in the data (Hyttinen, Eberhardt, and Järvisalo, 2014). Instead of building a graph directly, they test which variables stay dependent after conditioning on others, remove edges that cannot be direct, and orient the rest using graphical rules. The PC algorithm (Spirtes, Glymour, and Scheines, 1993) is the classic example, and

PCMCI+ extends this idea to multivariate time series by conditioning on the past of each series, which accounts for temporal dependence and autocorrelation (Runge, 2020; Runge, Nowack, et al., 2019). Their strengths are solid theory, interpretable graphs, and a good fit to observational data, which is why they are widely used in science. Their weakness is that they are only as good as the independence tests they rely on: the tests are sensitive to their assumptions, nonlinear dependencies need more complex tests, and small samples make the tests less reliable as the conditioning sets grow. These problems get worse in high dimensions, because the number of tests and their cost grow quickly with the number of variables (Runge, Gerhardus, et al., 2023).

Score-based methods frame causal discovery as a model-selection problem. They assign a score to each candidate graph, usually balancing how well the graph explains the data against how complex it is. The goal is then to find the graph with the best score. NOTEARS (Zheng et al., 2018) introduced an important version of this idea by formulating graph learning as a continuous optimization problem rather than a discrete search over possible graphs. DYNOTEARS (Pamfil et al., 2020) extends this approach to time-series data by allowing lagged dependencies between variables. These methods are flexible and avoid explicitly enumerating all possible graph structures, but they can still be computationally demanding and may become less reliable as the number of variables increases.

Functional causal models take a different approach. They infer causal direction from asymmetries in the data-generating process rather than from independence patterns or overall fit. They assume a specific functional relationship between cause, effect, and noise, and then pick the direction that stays consistent with the data. LiNGAM (Shimizu et al., 2006) does this for linear mechanisms with non-Gaussian noise, and additive noise models (Hoyer et al., 2008) extend it to nonlinear mechanisms, where the nonlinearity itself reveals the direction. The strength of this family is that it can tell cause from effect where correlation alone cannot, while keeping an interpretable mechanism. The cost is strong assumptions about the functional form and the noise, which may not hold in practice, and the models are hard to extend to complex, high-dimensional temporal systems.

All three families share a pattern. Each built part of the theoretical foundation of temporal causal discovery, but each also depends on assumptions to make discovery possible: reliable independence tests, a tractable global score, or a correct functional form. In real systems that are strongly nonlinear, noisy, high-dimensional, and full of long-range dependencies, these assumptions are increasingly hard to meet, and the guarantees behind each family weaken (Runge, Gerhardus, et al., 2023; Gong et al., 2023). This led

researchers to look for methods that model complex temporal dependencies directly from data, with fewer manual assumptions, and turned attention to representation learning and deep learning (Nauta, Bucur, and Seifert, 2019). The next section reviews these deep learning-based approaches.

2.3. Deep Learning-Based Temporal Causal Discovery

Deep learning offers another route: instead of fixing the form of the dependencies in advance, neural models learn them directly from data, which makes them attractive for nonlinear and long-range structures (Nauta, Bucur, and Seifert, 2019; Assaad, Devijver, and Gaussier, 2022). This does not remove the need for causal assumptions. It moves the burden from stated statistical conditions to the choice of architecture, inductive biases, optimization, and the amount and quality of training data. Deep learning-based causal discovery is therefore not assumption-free, it only changes where the assumptions sit (Gong et al., 2023; Peters, Janzing, and Schölkopf, 2017).

Early neural methods showed that networks can recover nonlinear temporal dependencies from observational data. An early and influential example is the Temporal Causal Discovery Framework (TCDF) (Nauta, Bucur, and Seifert, 2019). TCDF trains attention-based convolutional networks to predict each time series from the whole observed system, and reads the learned attention scores as signs of potential causal relevance. It then adds a permutation-based validation step, checking whether disrupting a candidate cause worsens the prediction, to separate mere predictive relevance from more plausible causal relations, and it also estimates the time delay between cause and effect. TCDF illustrates the shift from classical statistical methods toward neural architectures for temporal causal discovery. It is also directly relevant to this thesis, as it already shows that model-internal relevance scores, such as attention, are not enough on their own for causal interpretation and need a separate validation step. Other early neural approaches recovered nonlinear temporal dependencies in related ways, reformulating Granger causality with neural predictors (Tank et al., 2021) or learning latent interaction graphs from system trajectories (Kipf et al., 2018). Together, these methods made causal discovery more flexible by allowing nonlinear relationships to be learned directly from the data, rather than specified in advance by the researcher. Their weaknesses are also typical: results depend on architecture-specific inductive biases, the learned structure can be hard to interpret, capturing complex global dependencies is still difficult, and performance depends heavily on model design and the available training data.

Building on the move toward fully differentiable structure learning (Lachapelle et al., 2020; Gong et al., 2023), recent work has begun to explore transformer-based approaches to causal discovery. Attention is well suited to representing dependencies across a long temporal window and to modelling interactions both between variables and over time, which makes it a natural fit for multivariate temporal data (Shih, Sun, and Lee, 2019). Several recent works apply transformers to causal structure rather than to forecasting alone (Bi et al., 2023; Huang, Yao, and Divakaran, 2025). Naming can be a source of confusion here, since more than one model has been called a “CausalFormer” or similar. The framework of C. Zhang et al. (2023) explicitly builds on Granger-causality ideas in transformer forecasting. This differs from Kong et al. (2024), whose CausalFormer uses a prediction-trained causality-aware transformer and derives causal scores through model interpretation. Among these transformer-based approaches, the CausalFormer model of Kong et al. (2024) is the one used in this thesis. It is built for temporal causal discovery rather than forecasting, and it produces directed, lagged candidate edges between variables together with a mechanism that extracts this structure from the trained model. The authors report strong performance on synthetic, simulated, and selected real-world benchmarks. These properties fit the aim of this thesis, because the model produces exactly the kind of directed, lagged candidate relationships whose credibility can then be tested. Throughout, its outputs are treated as candidate causal structures that require validation, not as established causal facts.

Deep learning has therefore widened the set of tools for temporal causal discovery, extending it to the nonlinear, high-dimensional, and long-range dependencies that classical statistical methods struggle with. It has not, however, answered whether the discovered relationships are credible. These models are mostly evaluated on synthetic benchmarks with known ground truth, and in real observational settings a plausible, confidently produced graph may still reflect correlation, confounding, or preprocessing artefacts rather than a genuine causal structure (Assaad, Devijver, and Gaussier, 2022; Gong et al., 2023). The credibility of deep-learning-proposed causal edges therefore remains an open question. The next section turns to how the field evaluates causal discovery methods and identifies the gap this thesis addresses.

2.4. Evaluation and Validation of Temporal Causal Discovery

Evaluating a causal discovery method is harder than evaluating a forecasting model. In forecasting the target is observable, so accuracy can be measured against the realized values. In causal discovery the target is the causal graph, which in most real systems is never observed (Hasan, Hossain, and Gani, 2024; Moraffah et al., 2021). Evaluation therefore depends on whether ground truth is available. Synthetic and simulated systems can be built with a known graph, which a method’s output can then be compared against. Observational systems such as electricity markets give no such reference, only the recorded behaviour of the variables. This split between settings with and without ground truth shapes how the field evaluates temporal causal discovery (Assaad, Devijver, and Gaussier, 2022).

When a ground-truth graph is available, methods are judged by how closely the recovered graph matches it. Edge recovery is usually reported with precision, recall, and the F1 score, together with the structural Hamming distance, which counts the edge insertions, deletions, and reversals needed to turn the estimate into the truth; threshold-free measures such as the areas under the ROC and precision-recall curves are also common (Assaad, Devijver, and Gaussier, 2022). Temporal methods add one more requirement. A causal claim between time series concerns not only whether an edge exists, but also its direction and lag, so an evaluation that checks only edge existence says less than one that also checks direction and delay. CausalFormer, for example, is benchmarked on both edge recovery and lag recovery (Kong et al., 2024), and related work that probes temporal models for causal structure or extends constraint-based discovery to latent confounders is judged in the same way (Redden, Wang, and S. Li, 2026; Gerhardus and Runge, 2020).

These benchmarks are necessary but not sufficient. They are useful because they give known ground truth under controlled conditions, which allows fair comparison. But their value is only as good as the assumptions behind the data-generating process (Brouillard et al., 2025). Synthetic series can be too clean, too stationary, too small, or too closely matched to the assumptions of the algorithms being tested, and they often leave out the hidden confounding, nonstationarity, measurement noise, complex or multiple lags, and regime changes of real systems (Ferdous, Hossain, and Gani, 2025). A high F1 score or a low structural Hamming distance on such data shows that a method can recover

structure under those conditions. It does not show that the edges are credible in a real observational setting.

Aware of this gap, recent work has built semi-synthetic benchmarks generated from real recordings or detailed generative models (Cheng et al., 2024; Göbler et al., 2024) and large real-world benchmarks with curated reference graphs (Stein et al., 2025). These efforts make controlled evaluation more realistic, but they do not solve the evaluation problem for empirical domains such as electricity markets, where no complete reference graph is available and causal interpretations must still rely on assumptions, domain knowledge, and partial validation.

Where no ground-truth graph exists, evaluation cannot rely on recovery metrics, and the question changes from whether a graph is correct to whether its individual claims are credible (T. Li et al., 2024). Several ground-truth-free strategies address this. Falsification and cross-validation-style checks withhold part of the information, for example one variable, and test whether the discovered structure remains consistent. If a structure cannot be reproduced under this restriction, it is treated as less credible (Schkoda et al., 2024). Compatibility checks test whether a method produces consistent graphs when it is applied to overlapping subsets of the variables. If the same relation appears in two subsets, the method should infer it in the same way rather than reversing or contradicting it. Such checks therefore help identify structures that are unstable or unreliable (Faller et al., 2024). These are supported by robustness and stability checks across data subsets and by consistency with domain knowledge. None of these can prove that a relationship is causal. What they can do is give evidence against spurious structures and support edges that survive repeated testing (Faller et al., 2024).

The reviewed evaluation strategies show that there is no single standard for assessing temporal causal discovery. Synthetic benchmarks are useful when the true graph is known. Semi-synthetic and curated real-world benchmarks make this setting more realistic, but they still rely on some reference structure. In observational domains such as electricity markets, such a reference is usually not available. The central challenge is therefore not only whether a method performs well on benchmarks, but whether its individual discovered relations can be validated in a real empirical setting. This thesis addresses that challenge by combining benchmark evaluation with edge-level validation based on robustness, falsification, and domain plausibility.

2.5. Causal Analysis in Electricity and Energy Markets

Electricity markets are complex systems. Prices and system behaviour come from many interacting mechanisms such as demand, renewable generation, fuel and carbon prices, grid constraints, and cross-border market coupling (Lakić et al., 2025). Predictive models can find variables that help forecast prices, but a good predictor is not necessarily a causal driver, and correlation alone cannot say whether, or in which direction, one quantity affects another. Causal analysis is therefore needed to reason about mechanisms, direction, and price formation. It has become more relevant as the energy transition links renewables, demand, storage, and policy signals into a system that keeps changing (Ferkingstad, Løland, and Wilhelmsen, 2011; Cacciarelli et al., 2025).

Causal reasoning has a longer history in this field than the recent interest in causal discovery suggests. Ferkingstad, Løland, and Wilhelmsen (2011) study the causal relations between Northern European electricity prices and major fuel sources. They combine a vector error-correction model with LiNGAM to recover directed structure from observational data on Nordic and German electricity prices, oil, gas, coal, exchange rates, German wind power, and Nordic reservoir levels, and show how econometric modelling and non-Gaussianity can together identify directed relations in electricity and fuel markets. Castagneto-Gissey, Chavez, and De Vico Fallani (2014) build dynamic Granger-causal networks of European electricity spot prices, using multivariate Granger causality and network measures to track market integration and changing cross-border price links over time. Both show that causal connectivity has long mattered in electricity-market research, but both are statistical and econometric rather than deep-learning based, and neither looks at transformer-based discovery or at the edge-level validation used here.

A second strand focuses on renewables, whose effect on supply, price levels, and volatility is a central causal question. It helps to separate causal inference, which estimates the effect of a defined treatment, from causal discovery, which recovers structure. Cacciarelli et al. (2025) follow the first route, using double machine learning to estimate the causal effect of wind and solar generation on electricity prices, and find that these effects can be nonlinear rather than a simple uniform reduction, which is why a causal rather than a purely associational analysis is needed. Madadkhani et al. (2026) follow the second route, using LPCMCI to map the causal structure of price formation in Texas’s transitioning electricity market across electricity prices, hub differentials, wind and solar forecasts, gas generation and prices, load, and weather. This recent study is the closest precedent for the present work, since it uses causal discovery to find lagged parents of electricity prices, but it too uses a statistical discovery method rather than a transformer-based model.

Causal-network learning is also applied more broadly across energy-related markets. Recent work uses PCMCI-style discovery to trace direct and indirect, time-lagged paths by which attention to climate change moves through carbon, fossil-energy, and clean-energy markets (Liu, Tang, and Zhao, 2025), while the Texas study above shows LPCMCI applied under latent confounding and autocorrelation (Madadkhani et al., 2026). At the qualitative end, causal-loop modelling has been used to reason about feedback in energy-system adoption (Agnew, Smith, and Dargusch, 2018), though this is far from data-driven discovery. Together, these studies show that causal-graph reasoning and lagged causal networks have become common tools in energy research. They also share a limitation relevant here: their validation usually rests on statistical tests, robustness checks, historical events, or domain plausibility, not on a systematic, edge-level check of the candidate edges proposed by a learned model.

Overall, the energy-market literature shows that causal analysis is useful for understanding price formation, renewable integration, and interdependencies among electricity, fuel, carbon, and clean-energy markets. Transformer-based temporal causal discovery, however, has rarely been applied to electricity-market time series, and the directed, lagged edges such models propose have rarely been validated systematically on real observational data.

2.6. Research Gap and Positioning of this Thesis

The literature reviewed in this chapter points to a consistent picture. Temporal causal discovery is an active field because causal relations in observational time series are hard to identify. The classical statistical families, constraint-based, score-based, and functional, give firm theoretical foundations but rely on assumptions that become restrictive in nonlinear, noisy, and high-dimensional systems with long-range dependencies. Deep learning, and especially transformer-based models, extend what the field can model. Among them, CausalFormer (Kong et al., 2024) is directly relevant to this thesis, because it is built for temporal causal discovery and proposes directed, lagged candidate edges between variables. It is also methodologically relevant because it moves beyond earlier neural approaches that often derive causal relations from selected model components, such as attention scores or convolutional weights. Instead, CausalFormer uses a decomposition-based causality detector to derive causal scores from the trained transformer model more broadly.

However, this more complete model interpretation does not solve the validation problem. Two gaps remain. First, evaluation mostly uses synthetic and simulated benchmarks with known ground truth, while real observational systems, electricity markets among them, rarely offer a complete causal graph to check discovered edges against. Second, although causal analysis is by now well established in energy-market research, covering price formation, renewable integration, market coupling, and cross-market interdependencies, it has used almost only econometric and statistical causal discovery, causal inference with an assumed treatment, or qualitative causal modelling. The systematic validation of the directed, lagged edges produced by a transformer-based model, on real observational electricity-market data, is still largely unexplored.

This thesis addresses that gap. It applies CausalFormer to Germany-Luxembourg day-ahead electricity-market and weather time series, treats the resulting edges as candidate causal hypotheses rather than confirmed causal facts, tests them through a multi-stage validation framework, and interprets only the edges that pass in relation to electricity-market mechanisms. The contribution is therefore not a new causal discovery architecture, but a validation-centred empirical assessment of whether transformer-proposed causal edges are statistically credible and economically interpretable in a real electricity-market setting. The next chapter describes the data, the CausalFormer pipeline, and the validation framework used for this assessment.

3. Methodology

This thesis investigates whether CausalFormer (Kong et al., 2024), a transformer-based temporal causal discovery model, produces statistically credible and economically interpretable directed edges in the Germany-Luxembourg (DE-LU) day-ahead electricity market. The study is empirical and validation-centred: the primary contribution is not a new model architecture but a systematic assessment of whether transformer-proposed causal edges hold up under independent statistical scrutiny.

3.1. Research Design

CausalFormer is applied to the DE-LU electricity-market data to generate directed, lagged candidate causal edges between market and weather variables. These edges are treated as candidate causal hypotheses, not as established causal facts: observational time-series methods cannot establish causality in the absence of controlled experiments (Peters, Janzing, and Schölkopf, 2017). Each candidate is therefore evaluated through a multi-stage validation framework designed to filter spurious associations, and only the edges that survive this framework are interpreted in relation to electricity-market mechanisms.

Figure 3.1 summarises this workflow. The remainder of the chapter follows it: the data and its preprocessing, the CausalFormer pipeline that proposes the candidate edges, and the validation framework applied to them. The accepted edges are subsequently interpreted and discussed in the Results and Discussion chapters.

3.2. Data

The study focuses on the Germany-Luxembourg (DE-LU) electricity bidding zone. Seven time-series variables are analysed (Table 3.1). They were selected to capture the principal components of electricity-market dynamics: market outcomes (price), system demand (load), renewable generation (wind and solar), and the underlying meteorological drivers

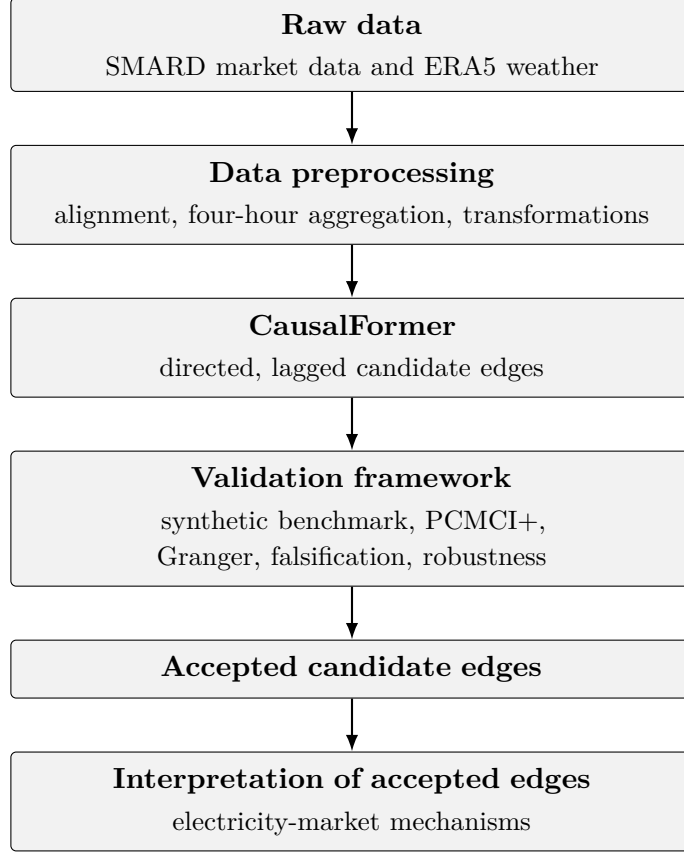


Figure 3.1.: Empirical workflow of the thesis. CausalFormer proposes directed, lagged candidate edges from the preprocessed DE-LU data; these are filtered by a multi-stage validation framework, and only the accepted edges are interpreted against electricity-market mechanisms.

(temperature, wind speed, and solar radiation). Together, these variables provide a compact representation of the key physical and economic processes expected to influence price formation.

The market variables are obtained from the SMARD transparency platform,¹ which publishes hourly metered data for the German bidding zone: the day-ahead electricity price, total system load, combined onshore and offshore wind generation, and photovoltaic solar generation. The weather variables are taken from the ERA5 reanalysis dataset (Hersbach et al., 2020), which provides gridded atmospheric fields at one-hour temporal and 0.25° spatial resolution: 2-metre air temperature, 10-metre wind speed, and surface downwelling shortwave solar radiation. Wind speed is derived as the Euclidean norm of the horizontal ERA5 wind components (u and v) computed per grid point before spatial averaging, so that opposing wind directions across the domain do not cancel. All weather variables are spatially averaged over the bounding box

¹www.smard.de, operated by the German Federal Network Agency (Bundesnetzagentur).

Table 3.1.: Variables used in the analysis.

Variable	Unit	Source	Role in the analysis
Price	EUR/MWh	SMARD	Target market outcome
Load	MWh	SMARD	Electricity demand
Wind generation	MWh	SMARD	Renewable supply
Solar generation	MWh	SMARD	Renewable supply
Temperature	°C	ERA5	Weather driver
Wind speed	m/s	ERA5	Weather driver
Solar radiation	W/m ²	ERA5	Weather driver

[47.27°N, 55.06°N, 5.87°E, 15.04°E], which encompasses the full German-Luxembourg territory.

The dataset covers 2019-2021, yielding 26 304 hourly observations. This three-year period spans diverse market conditions, including the COVID-19 demand shock of spring 2020 (International Energy Agency, 2020).

3.3. Preprocessing

Because the subsequent methods have different statistical assumptions and objectives, preprocessing is tailored to the requirements of each analysis rather than applying a single transformation across all of them.

Alignment and cleaning. All seven series are merged on a common UTC hourly index. Missing values are filled in using forward-filling up to three consecutive hours. Data integrity is verified by checking the expected row count (26 304 for 2019-2021) and plausible value ranges per variable (e.g. price in $[-500, 1000]$ EUR/MWh; wind speed in $[0, 40]$ m/s).

Temporal aggregation. Observations are aggregated to four-hour resolution by taking the within-block mean, reducing the 26 304 hourly observations to 6 576 four-hour periods. Hourly electricity-market variables exhibit strong autoregressive behaviour, meaning that recent values alone explain much of the current observation; during preliminary experiments this led CausalFormer to focus on self-dependencies, thereby suppressing the discovery of cross-variable relationships. Aggregating to four-hour resolution reduces

this short-term autocorrelation while preserving the dominant daily cycle. An empirical comparison of aggregation strategies is reported in Appendix A.

Deseasonalization. To remove systematic intraday and weekly periodicity before conditional-independence testing, a per-variable OLS regression is fitted with hour-of-day dummy variables (five indicators at four-hour resolution) and day-of-week dummies (six indicators), plus an intercept. Each observation is then replaced by its residual, the observed value minus the seasonal mean predicted by the regression, which gives the deseasonalized series. Annual seasonality is intentionally retained, as it carries economically meaningful signal.

First differencing. Aggregation weakens the autocorrelation but does not remove it entirely, so for CausalFormer the four-hour series are additionally first-differenced, $\Delta x_t = x_t - x_{t-1}$: each value is replaced by its change from the previous step. This further reduces residual autocorrelation and was adopted as the input representation for CausalFormer, selected empirically in comparison with the original and deseasonalized series (Appendix A). A known trade-off is that slowly varying physical couplings (for example, the level relationship between solar radiation and solar generation) may be attenuated in the differenced representation; this limitation is revisited in Section 5.4.

Matching representations to methods. Preprocessing produces three representations of the four-hour data, each chosen to best satisfy the assumptions of the method that receives it. The first-differenced series is the primary input to CausalFormer, where breaking the autoregressive shortcut is what allows cross-variable structure to surface. The deseasonalized series is used for PCMCI+ triangulation and the phase-randomization falsification test, both of which rest on conditional-independence testing and would otherwise be distorted by the shared seasonal patterns that deseasonalization removes. The original levels series is used for the time-shift falsification test and the seasonal regime-stability checks, which depend on the intact temporal ordering and seasonal structure that those procedures deliberately disrupt or exploit.

3.4. CausalFormer Pipeline

Model overview. CausalFormer (Kong et al., 2024) is a transformer-based architecture for temporal causal discovery. For it, prediction is only a means to an end. During training the model learns a sparse causal attention mask over the lagged input window,

which is then read out as a causal graph using relevance propagation. Each output edge encodes a directed, lagged relationship: variable j at lag ℓ is proposed as a cause of variable i at the current timestep.

Input representation. The model receives a sliding window of $T = 24$ timesteps (96 hours, approximately one day at four-hour aggregation) across all $N = 7$ variables. Values are normalized per variable to the range $[0.5, 1.0]$ internally prior to training. The output window length is fixed at $T - 1 = 23$, a structural constraint of the architecture.

Training configuration. The model is optimized with Adam (learning rate 10^{-3}) for up to 200 epochs with early stopping at patience 30. Attention sparsity is encouraged through an ℓ_1 regularization term (weight $\lambda = 10^{-10}$) applied to the mask entries, combined with an attention temperature $\tau = 100$ that sharpens the softmax distribution. The architecture uses a single transformer layer with embedding dimension $d_{\text{model}} = 256$, four attention heads, feedforward hidden size 256, and no dropout. A 10% holdout serves as the validation split for early stopping, and all stochastic operations use random seed 42.

Edge extraction. After training, candidate edges are extracted via regression relevance propagation (RRP), which back-propagates predictions through the attention and causal convolution layers to produce a relevance tensor $R \in \mathbb{R}^{N \times N \times T}$. The entry $R_{ij\ell}$ quantifies how strongly variable j at lag ℓ contributed to the model’s prediction for variable i . A KMeans clustering step with $n = 3$ clusters separates high-relevance edges from background; the top $m = 2$ clusters by mean relevance are retained, and the corresponding tuples (j, i, ℓ) form the candidate edge set. The original paper uses $n = 2$, $m = 1$ for simple synthetic structures (3-4 variables) but $n = 3$, $m = 2$ for more complex datasets such as Lorenz96 (10 variables); the latter configuration was adopted here as more appropriate for the 7-variable energy-market setting.

Self-loops. CausalFormer produces self-loop edges (i, i, ℓ) reflecting autoregressive dependencies within each series. These are expected and informative about the temporal structure of the data, but they cannot be validated by conditional-independence tests such as PCMCi+, which absorb self-influence into the conditioning set. Self-loops are therefore reported separately and excluded from the cross-variable validation pipeline.

3.5. Validation Pipeline

The validation pipeline has five checks, grouped into four stages, that each test a different aspect of edge credibility. The idea is triangulation: no single method can separate genuine causal structure from spurious associations in observational data, so each check targets a different way an edge can be misleading.

The synthetic experiments check whether CausalFormer can recover known structure at all, before any claim is made on real data. PCMCI+ is an independent conditional-independence check: it uses different assumptions than CausalFormer and does not depend on the transformer’s learned representations, so agreement between the two methods adds credibility, while disagreement is informative rather than final. Granger causality is a simple linear baseline: an edge with no linear predictive signal is unlikely to reflect a real dependency, whatever the transformer proposes. The falsification tests work like negative controls: they deliberately destroy the cross-series structure a real causal edge should depend on, and flag edges that still appear. The stability checks test whether accepted edges hold across data subsets, separating robust structure from artefacts of a particular time window or season. No single test is enough on its own; each stage is sensitive to a different source of spurious association.

Stage 1: Synthetic ground-truth recovery. Before applying CausalFormer to real data, its ability to recover known causal structure is verified on controlled synthetic experiments. The benchmark datasets are taken directly from the CausalFormer repository (Kong et al., 2024), ensuring that the evaluation uses the same data files as the original publication. Four ground-truth directed acyclic graphs (DAGs) are used: a diamond (four variables, two directed paths from a common cause to a common effect), a fork (one cause with two effects), a collider (two causes converging on one effect), and a mediator chain ($X_0 \rightarrow X_1 \rightarrow X_2$). Each DAG is instantiated in ten independent replicates; CausalFormer is retrained from scratch on each replicate using the same hyperparameter configuration as the original paper ($T = 16$, $m = 1$, $n = 2$).

Performance is evaluated with precision, recall, F1 score, structural Hamming distance (SHD), and probability of lag detection (PoD, the fraction of true edges for which the recovered lag matches the ground truth). These experiments calibrate expectations for the real-data application and provide evidence that the pipeline can recover known structure before any claims are made on observational data.

Stage 2a: PCMCI+ triangulation. As an independent, model-agnostic check, the PCMCI+ algorithm (Runge, 2020) is applied to the deseasonalized four-hour data. PCMCI+ is a constraint-based conditional-independence (CI) testing procedure that iteratively conditions on increasingly large parent sets to separate direct from indirect and spurious associations. The partial correlation (ParCorr) CI test is used with significance level $\alpha = 0.05$ and maximum lag $\tau_{\max} = 24$ timesteps, matching CausalFormer’s receptive field. For each CausalFormer cross-variable candidate edge $(j \rightarrow i, \ell)$, the corresponding PCMCI+ p -value and test statistic are recorded; an edge is considered PCMCI+-supported if $p < 0.05$.

Stage 2b: Granger causality baseline. Pairwise Granger causality tests (Granger, 1969) are conducted on daily-differenced data (shift of six timesteps, or 24 hours) using the chi-squared sum-of-squared-residuals (SSR) statistic at $\alpha = 0.05$ and maximum lag 24. Granger causality serves as a linear, bivariate baseline: if a CausalFormer edge has no Granger support, it is unlikely to reflect even a simple linear predictive association. Because pairwise tests on long real-world series frequently reach significance due to shared confounders, Granger support is treated as a complementary signal rather than a definitive criterion.

Stage 3: Falsification tests. Two negative-control experiments assess whether CausalFormer edges depend on genuine temporal alignment between series.

- **Time-shift placebo.** All non-price variables are shifted forward by 500 timesteps (≈ 83 days), breaking temporal alignment with the price series while preserving each individual series’ autocorrelation structure. CausalFormer is retrained on the disrupted data; any edge that reappears is flagged as potentially spurious.
- **Phase randomization.** The Fourier phases of the deseasonalized series, which encode how the variables align in time, are independently randomized per variable while the amplitudes are preserved. This destroys cross-series correlations while leaving each variable’s marginal distribution and autocorrelation structure intact. CausalFormer is retrained on the phase-randomized data; edges that reappear are flagged.

A candidate edge that reappears in either falsification test (that is, it appears even after genuine cross-series structure has been deliberately destroyed) is excluded from the validated set. An edge that disappears under both controls, on the other hand, gives

evidence that it depends on the genuine shared timing between the series rather than on properties of each series alone.

Stage 4: Stability checks. Two robustness tests assess whether edges are consistent across different subsets of the data.

- **Regime robustness.** CausalFormer is retrained separately on winter months (December-February, ≈ 540 timesteps) and summer months (June-August, ≈ 540 timesteps). Edges present in both seasonal subsets are flagged as regime-robust.
- **Block bootstrap.** Ten bootstrapped datasets are constructed by resampling contiguous two-week blocks (84 timesteps at four-hour resolution) with replacement, preserving the autocorrelation structure of the original series. CausalFormer is retrained on each sample, and a stability score is computed as the fraction of runs in which each edge appears. Edges with a stability score ≥ 0.6 (present in at least 6 of 10 bootstrap runs) are flagged as temporally stable.

Credibility criteria. The evidence from the validation stages is combined into a two-tier acceptance rule.

Tier 1: Accepted. An edge is accepted if neither falsification test reproduces it: it appears in neither the time-shifted nor the phase-randomized data. This indicates that the edge depends on the genuine temporal alignment between the series rather than on an artefact. PCMCI+ support is recorded for each edge but is treated as an informational signal rather than an acceptance requirement.

Tier 2: Strongly accepted. A Tier 1 edge is strongly accepted if it also satisfies at least one stability criterion: it is regime-robust (present in both the winter and summer splits) or has a bootstrap stability score of at least 0.6.

Edges that meet only some criteria are retained in the full evidence table and discussed qualitatively, so that partial support is visible rather than silently discarded.

Three caveats apply throughout. First, all data are observational: even an edge that passes every validation stage cannot be interpreted as proof of causality without experimental intervention (Peters, Janzing, and Schölkopf, 2017). Accepted edges are better understood as statistically supported predictive associations with plausible causal interpretations. Second, PCMCI+ uses a linear CI test (ParCorr), which may fail to detect nonlinear dependencies that CausalFormer’s attention mechanism can capture; low

PCMCI+ confirmation is therefore not automatically evidence against an edge, and may instead reflect that the two methods are sensitive to different kinds of dependency. Third, first-differencing weakens slowly varying level relationships (for example, the physical link between solar radiation and solar generation), which is a trade-off against the stationarity it provides. This limitation is revisited in Section 5.4.

4. Results

4.1. Synthetic Benchmark Performance

Before applying CausalFormer to observational electricity-market data, its ability to recover known causal structure was assessed on controlled synthetic benchmarks. Four ground-truth graph structures were used: diamond, fork, mediator chain, and collider (a v-structure). Each graph structure was evaluated on ten independently generated synthetic datasets with randomly drawn lags and additive Gaussian noise. This resulted in 40 experimental runs in total.

Table 4.1 reports precision, recall, F1 score, mean structural Hamming distance (SHD), and probability of correct lag detection (PoD) for each structure and overall.

Table 4.1.: Synthetic benchmark results (10 replicates per structure, seed 42). Metrics are averages across replicates. SHD counts edge insertions, deletions, and reversals relative to the ground-truth graph; lower is better. PoD is the fraction of true edges for which the recovered lag matches the ground truth.

Structure	Precision	Recall	F1	SHD	PoD
Diamond	0.933	0.537	0.671	4.10	0.602
Fork	0.835	0.620	0.704	2.40	0.375
Mediator	0.900	0.500	0.640	2.90	0.542
Collider	0.830	0.640	0.719	2.40	0.450
Overall	0.875	0.574	0.684	2.95	0.492

Across all 40 runs, CausalFormer achieved an overall F1 score of 0.684. Precision was consistently higher than recall, indicating conservative edge recovery: the model recovered only part of the true graph while producing relatively few false-positive edges. The mean SHD was 2.95. Lag detection was correct for approximately half of true edges (PoD = 0.492).

Table 4.2 compares the F1 and PoD results obtained here against the values reported in the original CausalFormer publication (Kong et al., 2024).

Table 4.2.: Comparison of F1 and PoD with the original CausalFormer paper. Paper values are means with standard deviations. Values in the “This work” columns are means across 10 replicates.

Structure	F1		PoD	
	This work	Paper	This work	Paper
Diamond	0.671	0.68 ± 0.08	0.602	0.74
Fork	0.704	0.79 ± 0.11	0.375	0.46
Mediator	0.640	0.71 ± 0.06	0.542	0.63
Collider	0.719	0.77 ± 0.05	0.450	0.59

The benchmark data files and hyperparameter configuration match the original paper ($T = 16$, $m = 1$, $n = 2$), allowing a close comparison with the published results. The F1 scores obtained fall below the published values for all structures: the differences in F1 were smaller than those in lag recovery. For the diamond structure, the measured F1 of 0.671 is within one standard deviation of the reported 0.68 ± 0.08 ; PoD gaps were larger, around 0.14 for the diamond and collider. Overall, graph recovery was stronger than lag recovery, and both remained below the published results. This gap is discussed in Section 5.4.

4.2. CausalFormer Candidate Graph

Applied to the DE-LU electricity-market data (2019-2021, four-hour resolution, first-differenced), CausalFormer proposed 24 directed edges: 7 self-loops and 17 cross-variable candidates. Each of the seven variables produced one autoregressive self-loop, with lags ranging from 1 to 6 timesteps (4-24 hours). As explained in Section 3.5, self-loops were excluded from the cross-variable validation pipeline.

The 17 cross-variable candidate edges and their proposed lags are shown in Figure 4.1 and listed in the full evidence matrix (Table B.1, Appendix B). These edges are model-generated candidates, not validated causal relationships. The remainder of this chapter reports how they performed under the validation framework.

4.3. Validation Results

Granger causality tests supported all 17 candidates ($p \approx 0$ for 16 edges; $p = 0.021$ for price $\rightarrow$ solar generation). PCMCI+ triangulation, applied to the deseasonalized data, found only 1 of 17 edges statistically significant at $\alpha = 0.05$: solar generation $\rightarrow$ solar

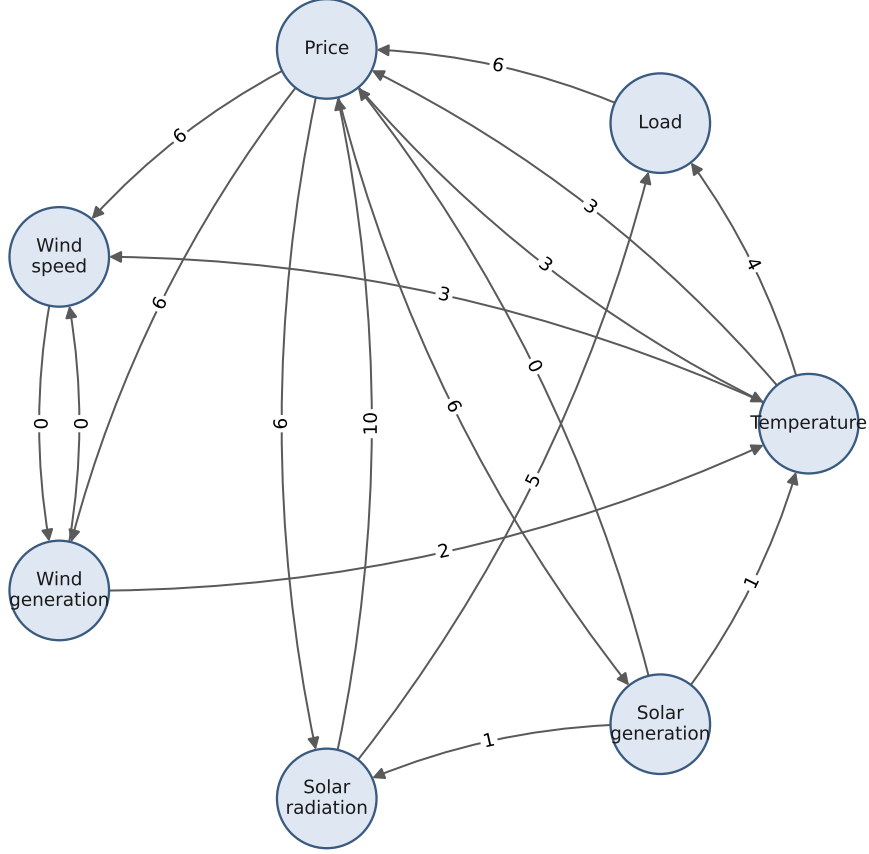


Figure 4.1.: Raw candidate graph proposed by CausalFormer for the DE-LU data (2019-2021, four-hour resolution, first-differenced). Nodes are the seven variables; directed edges are the 17 cross-variable candidates, labelled with the proposed lag in timesteps (one timestep is four hours). The seven autoregressive self-loops are omitted. These are model-generated candidates, before any validation.

radiation ($p = 0.007$). Granger causality therefore did not discriminate among the candidates, while agreement with PCMCI+ was rare.

The time-shift placebo flagged 7 edges and phase randomization flagged a separate set of 5. Because the two sets did not overlap, 12 of the 17 candidates were rejected under the falsification-based Tier 1 rule.

Among the 17 candidates, 11 achieved a block-bootstrap stability score of at least 0.6, meaning they appeared in at least 6 of 10 resampled datasets. Seven edges were regime-robust (present in both winter and summer splits). Five met both stability conditions. Five price-originating edges achieved bootstrap stability of 1.0 but were all flagged by the time-shift test (see Section 5.1).

Figure 4.2 summarizes this evidence for all 17 candidates across the six tests, ordered so the five accepted edges appear above the line.

4.4. Final Edge Classification

Five candidates met Tier 1 (not flagged by either falsification test); three of these also met Tier 2 (additionally bootstrap-stable or regime-robust). Table 4.3 lists the accepted edges, and Figure 4.3 shows them as a graph: of the 17 cross-variable candidates in Figure 4.1, only these five remain.

Table 4.3.: Accepted CausalFormer edges. ℓ = lag (in four-hour steps); Boot. = block-bootstrap stability score ($N = 10$); Reg. = regime-robust across seasonal splits. $\checkmark$ = yes; - = no; n/a = not applicable (concurrent edge at $\ell = 0$).

Cause	Effect	ℓ	Boot.	Reg.	PCMCI+ p	Tier
Tier 2: Strongly accepted						
Wind generation	Wind speed	0	0.7	-	n/a	T2
Temperature	Load	4	0.9	$\checkmark$	0.803	T2
Wind speed	Wind generation	0	0.9	-	n/a	T2
Tier 1: Accepted						
Solar generation	Temperature	1	0.0	-	0.150	T1
Solar radiation	Price	10	0.0	-	0.055	T1

Among the accepted edges, temperature $\rightarrow$ load is the only one that is both bootstrap-stable and regime-robust.

Two accepted edges, wind speed $\rightarrow$ wind generation and wind generation $\rightarrow$ wind speed, are contemporaneous (lag 0). CausalFormer can produce contemporaneous cross-variable edges because its architecture excludes instantaneous self-causality, but still allows lag-0 links between different variables. Since lag 0 provides no temporal ordering, the direction of such a link cannot be identified from the lag structure alone. It is therefore interpreted here as a contemporaneous association rather than as a directed causal effect. At the four-hour resolution used in this thesis, a lag-0 edge indicates co-movement within the same four-hour block and is therefore a weaker temporal claim than a lagged edge.

The complete evidence for all 17 candidates is reported in Table B.1 (Appendix B).

	PCMCI	Granger	Time shift	Phase rand.	Bootstrap	Season	Tier
Temperature → Load	X	✓	✓	✓	✓	✓	T2
Wind speed → Wind generation	–	✓	✓	✓	✓	X	T2
Wind generation → Wind speed	–	✓	✓	✓	✓	X	T2
Solar generation → Temperature	X	✓	✓	✓	X	X	T1
Solar radiation → Price	X	✓	✓	✓	X	X	T1
Price → Temperature	X	✓	X	✓	✓	X	–
Price → Wind speed	X	✓	X	✓	✓	X	–
Price → Solar radiation	X	✓	X	✓	✓	✓	–
Price → Wind generation	X	✓	X	✓	✓	✓	–
Price → Solar generation	X	✓	X	✓	✓	✓	–
Solar generation → Solar radiation	✓	✓	✓	X	✓	✓	–
Temperature → Price	X	✓	X	✓	✓	X	–
Temperature → Wind speed	X	✓	X	✓	✓	X	–
Load → Price	X	✓	✓	X	X	✓	–
Solar radiation → Load	X	✓	✓	X	X	X	–
Wind generation → Temperature	X	✓	✓	X	X	X	–
Solar generation → Price	–	✓	✓	X	X	✓	–

■ Passed / supportive
 ■ Failed / unsupportive
 ■ Not applicable

Figure 4.2.: Validation evidence for the 17 cross-variable candidates. Each row is a candidate edge and each column a validation test. Green marks supportive evidence (the edge passed the test), red marks failing or unsupportive evidence, and grey marks a test that does not apply (PCMCI+ does not test contemporaneous edges). For the falsification columns, passing means the edge was not recovered from the placebo data. The five accepted edges are ordered above the line, with their tier in the final column.

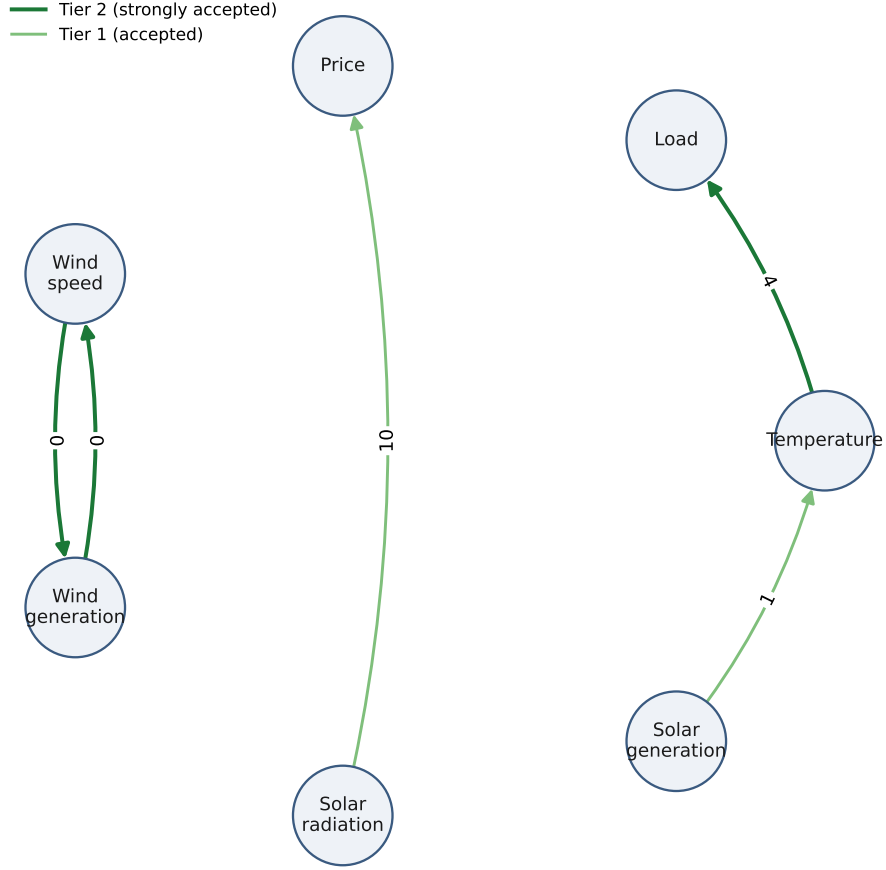


Figure 4.3.: Final accepted causal graph for the DE-LU data, using the same node layout as Figure 4.1. Only the five edges that passed the acceptance rule are shown, out of the 17 cross-variable candidates. Dark green marks the three Tier 2 (strongly accepted) edges; light green marks the two Tier 1 (accepted) edges. Edge labels give the lag in timesteps (one timestep is four hours).

5. Discussion

This chapter interprets the validation results, addresses the research questions, and discusses what the findings imply for the use of transformer-based causal discovery in electricity-market data.

5.1. Credibility of CausalFormer-Proposed Edges

The results for RQ1 suggest that CausalFormer’s proposed edges should be treated with caution unless they are independently validated.

The synthetic benchmarks in (Section 4.1) provide the first indication of this limitation. CausalFormer achieved moderate graph recovery, with an F1 score of 0.684, but lag recovery was weaker, with a PoD of 0.492. Its performance was also below the values reported in the original publication by Kong et al. (2024), with F1 differences of 0.05-0.10 and larger gaps in lag detection. The reason for this gap cannot be identified from the experiments alone. However, the result shows that reproducing transformer-based causal discovery performance is not straightforward.

The empirical validation on the DE-LU data points in the same direction. Of the 17 cross-variable candidates, 12 failed at least one falsification test. Validation was therefore not only a robustness exercise. Without it, most of the candidate set would include relationships that fail basic negative controls.

The price-originating edges illustrate the problem. CausalFormer proposed price as a cause of temperature, wind speed, solar radiation, wind generation, and solar generation. Each of these edges had a bootstrap stability score of 1.0. The three price-to-weather edges nevertheless fail a basic plausibility check. Since day-ahead electricity prices cannot affect weather conditions, their stability should not be interpreted as causal evidence. Instead, these edges indicate that statistically stable patterns can still be causally implausible. The price-to-generation edges are also difficult to interpret as direct causal mechanisms. In principle, market operation, redispatch, or curtailment could

create more complex feedback from prices to generation. However, no clear pathway of this kind is evident in the present results.

All five were flagged by the time-shift placebo. A model can be consistent and still wrong. Stability testing alone would have ranked these edges among the most trustworthy in the set. Only the falsification tests separated these stable but spurious edges from the rest.

The disagreement between CausalFormer and PCMCI+ is also substantial. Only one of the 17 candidates was confirmed by PCMCI+ at $\alpha = 0.05$, and that edge was itself rejected by phase randomization. None of the three Tier 2 edges received PCMCI+ support. One possible explanation is that the two methods use different data representations: PCMCI+ runs on deseasonalized data, while CausalFormer runs on first-differenced data. For temperature $\rightarrow$ load, deseasonalization may remove much of the daily and seasonal variation through which temperature and load move together, leaving a weaker residual relationship. The acceptance rule therefore treats PCMCI+ as an informational signal rather than a gate. Requiring its confirmation would have left almost no accepted edge. That absence of independent statistical support still weakens the overall evidence.

The overall answer to RQ1 is therefore cautious. CausalFormer’s proposed edges hold up under independent validation only to a limited degree. Only a minority of candidates survive the full validation procedure. Among them, the strongest evidence is found for physically plausible srelationships, while the weaker accepted edges remain uncertain.

5.2. Interpretation of the Accepted Dependencies

Which dependencies survived validation, and how should they be interpreted? This section addresses RQ2 by discussing the five accepted edges (Figure 4.3) in order of decreasing credibility.

Temperature to load. Temperature $\rightarrow$ load (lag 4, or 16 hours) is the strongest result. It passed both falsification tests, was bootstrap-stable (0.9), was regime-robust across the winter and summer splits, and has a direct physical mechanism. Electricity demand depends on temperature through heating and cooling load, a relationship that is well established in energy-market research (Ferkingstad, Løland, and Wilhelmsen, 2011).

The 16-hour lag is plausible but should not be interpreted too literally. Lag recovery was the weakest part of CausalFormer’s performance in the synthetic experiments ($\text{PoD} = 0.492$), so the method identifies an edge more reliably than its delay. The specific value

may instead reflect planning cycles, delayed heating or cooling response, or the four-hour aggregation, and may not be structurally meaningful. Among the discovered edges, this one provides the strongest causal hypothesis, although its exact lag and directness remain uncertain.

Wind speed and wind generation. Both directions between wind speed and wind generation were accepted at Tier 2. Wind speed $\rightarrow$ wind generation is physically obvious and bootstrap-stable (0.9). Wind generation $\rightarrow$ wind speed was accepted on the same terms (bootstrap 0.7). Both are contemporaneous edges ($\ell = 0$).

The validation framework cannot orient a contemporaneous association. The direction that makes physical sense comes from domain knowledge. Wind drives turbines, not the reverse. The reverse edge is best interpreted as the same contemporaneous association seen in the opposite direction. One plausible explanation is spatial aggregation. Both variables are measured as averages over the DE-LU area rather than as matched local wind-speed and turbine-output observations. Geographic aggregation is known to smooth wind-power variation and can weaken the direct correspondence between local wind conditions and local generation (Hasche, 2010). In this setting, the physically one-directional wind-to-power mechanism may therefore appear as a contemporaneous association that the model cannot reliably orient.

Weaker Tier 1 edges. Solar radiation $\rightarrow$ price (lag 10) and solar generation $\rightarrow$ temperature (lag 1) both passed falsification but appeared in none of the ten bootstrap resamples. They received negative-control support but no positive stability support, and should be treated as fragile.

Solar radiation $\rightarrow$ price is plausible as an indirect pathway: radiation drives solar generation, which lowers price through the merit-order effect. However, the 40-hour lag does not fit a same-day mechanism. Given the weak lag recovery in the synthetic experiments, this lag is likely an artefact rather than a meaningful delay.

Solar generation $\rightarrow$ temperature is implausible in its stated direction. Solar generation cannot plausibly cause regional air temperature at the scale studied here. The edge more likely reflects a shared meteorological driver, particularly solar radiation, or remaining seasonal dependence. It illustrates that formal acceptance does not imply a convincing causal interpretation.

Overall, the accepted set contains one comparatively strong causal hypothesis (temperature $\rightarrow$ load), one robust but directionally unresolved contemporaneous relationship (wind speed and wind generation), and two fragile Tier 1 candidates.

5.3. Methodological Implications

Three general lessons emerge from applying a validation framework to transformer-generated causal graphs.

First, model output requires external validation. CausalFormer produced a confident graph, but confidence is not credibility. The price-originating edges (Section 5.1) demonstrate that repeated recovery measures model consistency rather than causal validity. Falsification tests provide a different kind of evidence than stability tests, and neither alone would have removed all spurious candidates.

Second, different validation checks provide complementary evidence. The time-shift and phase-randomization tests flagged non-overlapping sets of edges. Granger causality supported all candidates and had no discriminatory value. A multi-method approach is more informative than any single test.

Third, statistical validation cannot replace domain knowledge or establish causality. For contemporaneous edges, the correct direction comes from domain knowledge. Recognizing that solar generation $\rightarrow$ temperature is implausible requires knowing how weather and generation are physically related. The main value of the validation framework was not confirming a large causal graph; it was reducing a confidently produced graph to a smaller set of more credible hypotheses.

5.4. Limitations

Several limitations affect the conclusions of this study.

Fundamental limits. The data are observational. No intervention was performed, and no causal effect can be established with certainty (Peters, Janzing, and Schölkopf, 2017). Unlike the synthetic experiments, the DE-LU data have no ground-truth causal graph. Edges that pass all tests may still be spurious, and edges that fail may reflect real but undetected relationships. The acceptance criteria were designed for this study and were not externally calibrated; Tier 1 and Tier 2 represent relative evidence within the framework rather than universally calibrated credibility levels.

Data scope. The seven variables omit important drivers of electricity prices: fuel prices, cross-border flows, installed capacity, forecasts, and grid constraints. Omitted confounders may explain some of the accepted edges or mask others. Weather variables are spatially averaged over the DE-LU bidding zone, which may hide regional relationships. The study

period (2019–2021) spans the COVID-19 demand shock of spring 2020, introducing a structural break that is not explicitly modelled.

Preprocessing. Aggregation and differencing change the dependencies the model can detect. The choice of four-hour aggregation and first differencing was motivated by exploratory experiments (Appendix A), but different choices might yield different results. A systematic sensitivity analysis was not conducted.

Validation design. The time-shift falsification test shifts all non-price variables together, which breaks their alignment with price but preserves alignment among the non-price variables themselves. As a result, the test can only validly flag edges involving price. Edges between two non-price variables (such as the three Tier 2 edges) were not tested by the time-shift design. Each validation test uses the data representation appropriate to its purpose. CausalFormer, Granger, and the time-shift test run on first-differenced data. PCMCI+, bootstrap, and phase randomization run on deseasonalized data. Regime stability runs on the original levels. Disagreement between tests may therefore reflect either spuriousness or representation sensitivity. PCMCI+ uses a linear partial-correlation test, which cannot detect nonlinear dependencies. Only ten bootstrap runs were used. A larger number might produce more reliable stability estimates.

Method limitations. Synthetic lag recovery was weaker than graph recovery ($\text{PoD} = 0.492$). The specific lags proposed for the DE-LU edges may not be structurally meaningful. The validation framework cannot orient contemporaneous edges; both directions between wind speed and wind generation were accepted, although only wind speed $\rightarrow$ wind generation has a plausible direct mechanism at the spatial scale studied. The synthetic benchmark performance fell below the original CausalFormer publication, and the cause of this gap is not identified.

5.5. Implications for Future Research

Three directions follow from the limitations above.

First, replication across time periods and bidding zones would test whether the accepted edges generalize. Including additional variables such as fuel prices and cross-border flows would address the omitted-confounder limitation and test whether the accepted edges remain stable.

Second, methodological refinements could strengthen the validation. Nonlinear conditional-independence tests might improve agreement between CausalFormer and PCMCI+. Larger bootstrap samples would produce more reliable stability estimates. A systematic preprocessing sensitivity analysis would clarify how robust the findings are to aggregation and differencing choices.

Third, investigating open questions would clarify the scope of transformer-based causal discovery. Methods that can orient contemporaneous edges would address the bidirectional wind-speed result. Investigating the gap between the original CausalFormer results and this replication would clarify whether the issue is environmental, methodological, or data-dependent.

6. Conclusion

This thesis set out to address a validation problem. Transformer-based causal discovery produces directed, lagged candidate edges, but model-derived relevance scores are not causal evidence on their own, and real electricity-market data offer no ground truth against which to check the result. CausalFormer was applied to Germany-Luxembourg electricity-market and weather data from 2019 to 2021, and the candidate edges it proposed were not treated as causal facts. Instead, they were tested with a multi-stage validation framework to assess whether they were statistically credible, robust, and physically or economically interpretable to different degrees.

The answer to RQ1 is that the raw CausalFormer graph receives only limited support under independent validation. The synthetic benchmark already showed moderate graph recovery but weaker lag recovery, so even the edges the model finds come with unreliable delays. On the real data, most candidate edges did not survive falsification. Stability alone was not enough. Several edges that were highly stable under bootstrap resampling were physically implausible, such as price appearing to drive the weather, and only the falsification tests separated these from credible candidates. A confident model output is therefore not the same as a credible one.

For RQ2, only a small subset of edges received enough support to be read as candidate causal hypotheses. The strongest was temperature to load, which passed both falsification tests, was stable under bootstrap resampling, was robust across seasons, and has a well-understood mechanism through heating and cooling demand. The relationship between wind speed and wind generation was also robust, but it is contemporaneous, so the validation framework cannot assign its direction. Domain knowledge, not the data, indicates that wind drives generation. The two weaker Tier 1 solar edges passed falsification but lacked stability support and remain physically uncertain. The validation-supported set is small, and its members differ in how much trust they warrant.

The main contribution of this thesis is the validation framework itself rather than any single edge. It combines synthetic benchmarking, statistical triangulation, falsification, and robustness checks, and it is the combination that carries the weight. No single test was sufficient. Falsification caught stable but spurious edges that stability alone would

have endorsed, while the stability checks separated robust survivors from fragile ones. Applied together, they turn a confident but unverified graph into a small set of edges that are credible under the framework, each with an explicit level of support.

These conclusions are bounded by the study design. The data are observational and cover one bidding zone over three years, the variable set omits drivers such as fuel prices and cross-border flows, and the results depend on preprocessing choices that were not varied systematically. The accepted edges are credible within this framework and this dataset, not established causal facts.

The broader implication is that transformer-based causal discovery is useful as a hypothesis generator, not as a source of causal conclusions. On real observational energy-market data, CausalFormer proposed many relationships, but most did not withstand independent scrutiny, and the ones that did still required domain knowledge to interpret. Placed between the model and any causal claim, independent validation and domain reasoning let these methods point to where a relationship may exist. Without that step, their outputs are candidate patterns rather than evidence of cause.

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A. Preprocessing Selection Experiments

The preprocessing choices described in Section 3.3 were selected through a sequence of exploratory experiments rather than adopted by default. This appendix documents those experiments and the reasoning behind the final configuration. They were carried out during a development phase on a 2021-2022 hourly window (17 520 observations) and used untuned configurations; the resulting strategy was then applied to the final 2019-2021 dataset. The edge counts reported here are therefore developmental evidence that motivated the configuration, not validated findings, and should be read as such.

A.1. Autoregressive dominance

Hourly energy series are highly persistent: the lag-1 autocorrelation of the seven variables ranges from 0.957 to 0.996, so each hour’s value is almost identical to the previous one. CausalFormer discovers structure indirectly, by training a predictive model and then attributing its predictions back to the input variables. When a variable can be predicted almost perfectly from its own immediate past, the model has no incentive to use any other variable, and the recovered graph collapses to autoregressive self-loops. This is a poor match for the data on which CausalFormer was originally evaluated, which had low autocorrelation (synthetic additive-noise models, the chaotic Lorenz96 system, and fMRI signals with lag-1 autocorrelation of roughly 0.3 to 0.5). The original paper does not address strong autocorrelation.

A.2. Temporal resolution

Three resolutions were compared by inspecting the relevance each variable assigned to its own past (self-relevance) and the cross-variable signal that remained, holding the clustering step fixed at its most conservative setting (Table A.1). On raw hourly levels, self-relevance was 84 to 96% per variable and no cross-variable edge was recovered. First differencing, the standard statistical remedy for non-stationarity, reduced self-

relevance only modestly, to 74 to 88%, and still produced no cross-variable edge. The limited effect has a physical explanation: the persistence of energy series is not a unit root but the smoothness of weather and demand processes, so the first differences are themselves persistent. Aggregating to four-hour means reduced self-relevance much further, to 35 to 73%, and recovered the first cross-variable edge (temperature to load) even under conservative clustering. Four-hour aggregation was therefore adopted as the base resolution.

Table A.1.: Effect of temporal resolution on autoregressive dominance and edge recovery (exploratory runs, conservative binary clustering). Self-relevance is the share of relevance a variable assigns to its own past, reported as the range across the seven variables.

Resolution	Self-relevance	Cross-variable edges	Core edge recovered
Raw hourly	84-96%	0	none
First-differenced	74-88%	0	none
Four-hour mean	35-73%	1	temperature to load

A.3. Input representation

On the four-hour data, three representations were compared as input to CausalFormer: the original levels, a deseasonalized version, and a first-differenced version (Table A.2). Deseasonalization removed the seasonal mean but left the residuals persistent, and CausalFormer again recovered only self-loops, with no cross-variable edge. The original levels produced cross-variable edges but also physically impossible ones, in which price appears to drive weather, reflecting the shared seasonal cycle. First differencing broke the autoregressive shortcut and produced cross-variable structure without removing the seasonal component, and was adopted for the final run.

Deseasonalization was therefore reserved for the validation stage rather than the discovery stage, because the two transformations address different problems. Differencing removes the persistence that lets CausalFormer rely on each variable’s own past, whereas deseasonalization removes the shared seasonal cycle that produces spurious co-movement between variables. Removing the seasonal component before discovery is counterproductive, because some genuine causal effects act through the daily cycle: solar radiation drives solar generation precisely because both follow the sun, and removing that cycle would erase the signal. The seasonal confound is instead removed downstream, where PCMCI+ tests whether an edge survives once the shared cycle is conditioned away. This

division of labour is why the two methods operate on different representations, a choice whose cost, reduced comparability between CausalFormer and PCMCI+, is discussed in Section 5.4. Applied to the final dataset, four-hour aggregation followed by first differencing produced the 24 edges analysed in Chapter 4.

Table A.2.: Candidate input representations for CausalFormer on the four-hour data (exploratory runs, ternary clustering). Edge counts are illustrative of the qualitative outcome rather than a controlled comparison.

Representation	Cross-variable edges	Self-loops	Notable outcome
Original levels	15	7	physically impossible price-to-weather edges
Deseasonalized	0	7	only self-loops recovered
First-differenced	11	6	cross-variable structure present; adopted

B. Full Validation Evidence Table

Table B.1 reports the complete validation evidence for all 17 cross-variable candidate edges proposed by CausalFormer. Columns indicate: the proposed lag (ℓ); the PCMCI+ p -value (n/a for concurrent edges at $\ell = 0$); Granger support (all edges supported at $\alpha = 0.05$); whether the edge was flagged as potentially spurious by the time-shift (TS) or phase-randomization (PR) falsification tests; whether the edge was regime-robust across seasonal splits (Reg.); the block-bootstrap stability score (Boot., fraction of 10 resampled runs in which the edge appeared); and the final acceptance tier. Self-loop edges are excluded.

Table B.1.: Complete validation evidence for all 17 cross-variable CausalFormer candidate edges. TS = time-shift falsification flag; PR = phase-randomization falsification flag; Reg. = regime robustness; Boot. = block-bootstrap stability score ($N = 10$). ✓ = yes/supported/flagged; - = no/not supported/not flagged. * $p < 0.05$ (PCMCI+). Rows sorted by acceptance tier.

Cause	Effect	ℓ	PCMCI+ p	Granger	TS	PR	Reg.	Boot.	Tier
Tier 2: Strongly accepted									
W. Gen.	W. Speed	0	n/a	✓	-	-	-	0.7	T2
Temp.	Load	4	0.803	✓	-	-	✓	0.9	T2
W. Speed	W. Gen.	0	n/a	✓	-	-	-	0.9	T2
Tier 1: Accepted									
S. Gen.	Temp.	1	0.150	✓	-	-	-	0.0	T1
S. Rad.	Price	10	0.055	✓	-	-	-	0.0	T1
Not accepted: falsification flag or insufficient stability									
Price	Temp.	3	0.316	✓	✓	-	-	1.0	-
W. Gen.	Temp.	2	0.290	✓	-	✓	-	0.0	-
Temp.	W. Speed	3	0.271	✓	✓	-	-	0.8	-
Price	W. Speed	6	0.958	✓	✓	-	-	1.0	-
Price	S. Rad.	6	0.083	✓	✓	-	✓	1.0	-
S. Gen.	S. Rad.	1	0.007*	✓	-	✓	✓	0.9	-
Temp.	Price	3	0.070	✓	✓	-	-	0.9	-
Load	Price	6	0.798	✓	-	✓	✓	0.2	-
S. Gen.	Price	0	n/a	✓	-	✓	✓	0.0	-
S. Rad.	Load	5	0.092	✓	-	✓	-	0.1	-
Price	W. Gen.	6	0.840	✓	✓	-	✓	1.0	-
Price	S. Gen.	6	0.409	✓	✓	-	✓	1.0	-

Abbreviations: Temp. = temperature; W. Speed = wind speed; W. Gen. = wind generation; S. Gen. = solar generation; S. Rad. = solar radiation. Granger causality was supported for all 17 edges ($p \approx 0$, except Price $\rightarrow$ S. Gen. where $p = 0.021$); the column is retained for completeness.